Drowsiness Detection System

An Engineering Project in Community Service

Phase – II Report

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Bonafide Certificate

Certified that this project report titled **Drowsiness Detection System** is the bonafide work of "23BCE10958 Ishit Chhabra, 23BCE10295 Kanak Holkar, 23BCE11086 Rohini Khapne, 23BCE10222 Kopal Verma, 23BAI10453 Harsh Rathore, 23BCY10202 Ishika Thakor" who carried out the project work under my supervision.

This project report (Phase II) is submitted for the Project Viva-Voce examination held on 30/03/2026.

Supervisor

Comments & Signature (Reviewer 1)

Comments & Signature (Reviewer 2)

Declaration of Originality

We, hereby declare that this report entitled **Drowsiness Detection System** represents our original work carried out for the EPICS project as a student of VIT Bhopal University and, to the best of our knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution. Works of other authors cited in this report have been duly acknowledged under the section "References".

30/03/2026

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This project was completed through the collective efforts of our team of six members. We sincerely appreciate the cooperation, coordination, and teamwork demonstrated by each member, which played a crucial role in the successful implementation of this system. Finally, we would like to thank our friends and family for their support and motivation throughout the project duration.

Abstract

Driver drowsiness is a major contributor to road accidents worldwide, significantly impairing reaction time, decision-making ability, and overall driving performance. Addressing this critical safety issue, this project presents a real-time **Drowsiness Detection System** that utilizes computer vision techniques to monitor driver alertness and provide timely warnings.

The system is implemented using Python and OpenCV, employing Haar Cascade classifiers for efficient face and eye detection. It continuously analyzes live video feed from a standard webcam to identify two primary indicators of fatigue: prolonged eye closure and yawning. If the driver's eyes remain closed for more than **2 seconds** or if yawning persists for more than **1 second**, the system classifies the driver as drowsy. To ensure reliability, frame-based temporal filtering is applied, allowing the system to distinguish between normal blinking and dangerous fatigue conditions.

A multi-modal alert mechanism is integrated to enhance responsiveness. When drowsiness is detected, the system activates an **audio alert** using a continuous 880 Hz tone along with **visual warnings**, including a flashing red border and on-screen status messages. The alerts automatically stop once the driver regains alertness. The entire process operates in real time with minimal latency (less than 100 ms), achieving processing speeds of approximately 20–30 frames per second on standard consumer hardware.

One of the key advantages of the proposed system is its lightweight and privacy-preserving design. Unlike advanced driver monitoring systems that require specialized hardware or cloud-based processing, this solution operates entirely on local devices without storing or transmitting any personal data. Experimental results demonstrate an overall accuracy of approximately 91% under normal lighting conditions, with low false positive rates.

In conclusion, the proposed Drowsiness Detection System provides a cost-effective, non-intrusive, and efficient solution for enhancing road safety. It demonstrates the practical applicability of classical computer vision techniques in real-world scenarios and highlights the potential for further improvements through integration with advanced machine learning models.

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1. INTRODUCTION

Road safety has become a critical global concern due to the rapid increase in the number of vehicles and the growing demand for long-distance and continuous driving. Among the various causes of road accidents, driver drowsiness is one of the most dangerous and often underestimated factors. Fatigue significantly impairs a driver's reaction time, decision-making ability, concentration, and situational awareness. In many cases, drowsiness leads to microsleep episodes—brief, involuntary periods of sleep—that can result in severe accidents, injuries, and fatalities. Therefore, early detection of driver fatigue and timely alert mechanisms are essential for improving road safety.

Driver drowsiness is typically characterized by behavioral and physiological indicators such as prolonged eye closure, frequent blinking, yawning, and reduced facial activity. Traditional methods for detecting fatigue include physiological monitoring techniques such as Electroencephalography (EEG), heart rate monitoring, and wearable sensors. Although these approaches provide high accuracy, they are often intrusive, expensive, and impractical for everyday use. In contrast, computer vision-based systems offer a non-intrusive, cost-effective, and real-time alternative by analyzing facial features through a standard camera.

This project presents a real-time Drowsiness Detection System developed using computer vision and image processing techniques. The system utilizes a standard webcam to continuously monitor the driver's facial features and detect signs of fatigue. It focuses on two primary indicators: prolonged eye closure and yawning. The detection mechanism is implemented using Haar Cascade classifiers in OpenCV for face and eye detection, which are efficient and suitable for real-time applications with limited computational resources.

The system operates by first detecting the driver's face in each video frame and then identifying the eye regions within the detected face. If the eyes remain closed for a continuous duration exceeding a predefined threshold (approximately 2 seconds), the system interprets it as a sign of drowsiness. Additionally, the mouth region is analyzed using texture variation and intensity-based features to detect yawning. If yawning persists beyond a specified time threshold (approximately 1 second), the system triggers an alert. This dual-indicator approach improves the reliability of fatigue detection compared to single-parameter systems.

Upon detecting drowsiness, the system activates both audio and visual alert mechanisms. An audible warning is generated to immediately capture the driver's attention, while visual cues such as a flashing border and on-screen status messages reinforce the alert. These alerts continue until the driver returns to an alert state. The entire process is executed in real time, ensuring minimal delay between detection and response.

The proposed system is implemented using Python along with libraries such as OpenCV for image processing, NumPy for numerical computations, and Pygame for audio generation. One of the key advantages of this system is its privacy-conscious design, as all processing is performed locally without storing or transmitting any video data.

This project is made efficient and accessible, and by leveraging classical computer vision techniques, the system provides a cost-effective solution for detecting driver fatigue and enhancing road safety.

1.1 MOTIVATION

Road safety has become an increasingly critical concern in modern society due to the rapid growth in the number of vehicles and extended driving durations. Among the various causes of road accidents, driver fatigue and drowsiness remain one of the most dangerous yet often overlooked factors. Drowsiness significantly reduces a driver's alertness, slows reaction time, and impairs decision-making ability. Unlike distractions such as mobile phone usage, fatigue develops gradually, making it difficult for drivers to recognize the onset of sleepiness. This often leads to microsleep episodes, which can result in severe accidents and loss of life.

The growing number of fatigue-related accidents highlights the urgent need for an efficient and real-time driver monitoring solution. Traditional methods for detecting drowsiness, such as Electroencephalography (EEG), heart rate monitoring, and wearable sensors, offer high accuracy but are impractical for everyday use due to their intrusive nature, high cost, and user discomfort. This creates a gap for a non-intrusive, affordable, and scalable solution that can be widely adopted.

With advancements in computer vision and image processing, it is now possible to monitor human facial behavior in real time using standard cameras. This project is motivated by the idea of leveraging these technologies to develop a practical and accessible system that can detect early signs of driver fatigue. By focusing on visual indicators such as prolonged eye closure and yawning, the system provides a reliable method for assessing driver alertness without requiring specialized hardware.

Another key motivation behind this project is to develop a cost-effective solution that can be implemented using commonly available devices such as laptops or webcams. Most advanced driver monitoring systems are integrated into high-end vehicles and rely on expensive sensors or proprietary technologies. In contrast, this system demonstrates that effective drowsiness detection can be achieved using open-source tools and lightweight algorithms, making it suitable for wider community use.

From an academic perspective, this project provides an opportunity to apply theoretical knowledge of computer vision to a real-world problem. It involves concepts such as face detection, feature extraction, real-time video processing, and threshold-based decision-making. Additionally, the project encourages the development of problem-solving skills, system design thinking, and teamwork, as it requires integration of multiple modules into a cohesive system.

Ultimately, the motivation for this project extends beyond technical implementation. It aims to contribute toward improving road safety and reducing accidents caused by driver fatigue. By developing a real-time Drowsiness Detection System, this project reflects the application of engineering principles for societal benefit, aligning with the core objectives of community-oriented engineering initiatives.

1.2 OBJECTIVE

The primary objective of this project is to design and develop a real-time Drowsiness Detection System that enhances road safety by identifying early signs of driver fatigue and providing immediate alerts. Driver drowsiness is a significant cause of road accidents, and timely detection can help prevent potentially life-threatening situations.

The system aims to utilize computer vision techniques to continuously monitor the driver's facial features using a standard camera. By analyzing key indicators such as eye closure duration and yawning behavior, the system determines the driver's level of alertness and responds accordingly. The goal is to create a non-intrusive and efficient monitoring solution that does not require wearable devices or specialized hardware.

A key objective is to ensure real-time performance, where video frames are processed continuously with minimal delay, allowing the system to respond instantly when drowsiness is detected. The system is designed to operate with low computational requirements so that it can run on standard consumer devices such as laptops without the need for high-end GPUs.

Another important objective is to implement a reliable threshold-based detection mechanism. The system tracks eye closure and mouth opening over time and triggers alerts only when predefined thresholds are exceeded. This approach helps distinguish between normal behaviors such as blinking and actual signs of fatigue, thereby reducing false alarms.

The project also focuses on developing an effective alert mechanism that includes both audio and visual feedback. The audio alert is generated as a continuous tone to capture the driver's attention, while visual indicators such as warning messages and flashing borders provide additional awareness.

Furthermore, the system is designed with a strong emphasis on privacy and security. All processing is performed locally on the device, and no video data is stored or transmitted, ensuring user confidentiality.

In addition to technical objectives, this project aims to demonstrate the practical application of computer vision in solving real-world problems. It also serves as a platform for learning concepts such as real-time image processing, feature detection, and system integration.

Overall, the objective of this project is to develop a cost-effective, reliable, and user-friendly drowsiness detection system that contributes to safer driving practices and reduces fatigue-related road accidents.

2. EXISTING WORK/LITERATURE REVIEW

Author	Title	Journal	Methodologies	Research Gap
1. Yaman Albadawi , 2. MaenTakruri , 3. MohammedAwad 2022	A Review of Recent Developments in Driver Drowsiness Detection Systems	*PeerJ* Sensors	The study proposes a real-time driver drowsiness detection system using computer vision techniques. It utilizes facial feature extraction through OpenCV and Haar Cascade classifiers to detect eye closure and yawning. The system monitors eye aspect ratio and mouth region behavior through continuous frame analysis and triggers alerts when predefined thresholds are exceeded	• Limited robustness under poor lighting and varied head positions. • Performance may decrease with occlusions such as glasses or masks. • Lacks advanced deep learning integration for higher accuracy and adaptability.
1. **Jagbeer Singh** 2. **Ritika Kanojia** 3. **Rishika Singh** 4. **Rishita Bansal** 5. **Sakshi Bansal** **2022**	Driver Drowsiness Detection System – An Approach By Machine Learning Application	Journal of Pharmaceutical Negative Results, Volume 13, Special Issue 10, 2022	The study proposes a machine learning–based driver drowsiness detection system using facial landmark detection. It applies OpenCV and Dlib's 68 facial landmark model to compute Eye Aspect Ratio (EAR) and PERCLOS for eye state analysis. A webcam captures real-time video, and a scoring mechanism triggers an alarm when eye closure exceeds a predefined threshold. The system was tested under different lighting conditions and driver variations.	• Accuracy affected by poor lighting and night conditions. • Fixed EAR threshold varies across individuals. • Limited robustness under sunglasses and environmental variations. • Achieved moderate accuracy (~80%), indicating scope for improvement using advanced deep learning models.
Islam A. Fouad **2023**	Driver Drowsiness Detection Using Convolutional Neural Networks	International Journal of Engineering Research & Technology (IJERT)	• Image-based driver monitoring using camera input. • Face and eye region detection using Haar Cascade classifier. • Convolutional Neural Network (CNN) used for classifying eye state (open/closed). • Dataset of eye images used for training and testing. • Real-time prediction of drowsiness based on continuous eye closure. • Alert system activated when drowsiness detected.	• Performance decreases in low-light and night driving conditions. • Sensitive to head pose variations and occlusions. • Limited dataset size affects generalization. • Does not integrate multiple fatigue indicators (yawning, head nodding). • Needs higher accuracy for real-world deployment.
1. **Vinu Sherimon** 2. **Sherimon** 3. **Alaa Ismaeel** 4. **Alex Babu** 5. **Sajina Rose** 6. **Sarin Abraham** 7. **Johnsymol Joy** **2023**	Real-Time Driver Drowsiness Detection Using Eye Blink Monitoring	International Journal of Recent Technology and Engineering (IJRTE)	• Real-time video capture using webcam. • Face detection using Haar Cascade classifier. • Eye region extraction and blink detection algorithm. • Eye Aspect Ratio (EAR) used to measure eye closure duration. • Threshold-based system to detect prolonged eye closure. • Alarm system activated when drowsiness level exceeds limit. • Implemented using Python and OpenCV for real-time processing.	• Fixed threshold may not work equally for all drivers. • Accuracy affected by lighting variations and camera angle. • Does not detect yawning or head movement. • Limited testing in real driving environments. • Needs improvement for higher robustness and real-world application.

Author	Title	Journal	Methodologies	Research Gap
1. **Fogue et al.** **2024**	Automatic Accident Detection: Assistance Through Communication Technologies and Vehicles	IEEE Vehicular Technology Magazine, 2012	The study proposes a smartphone-based accident detection system using the device's built-in accelerometer and GPS. It detects crashes by analyzing sudden deceleration patterns and automatically sends SMS alerts with GPS coordinates to emergency responders and predefined contacts. The system relies entirely on smartphone sensors and mobile networks for communication	• Fully dependent on smartphone availability and proper placement inside the vehicle. • No integration with vehicle hardware or additional sensors for improved accuracy. • Lacks driver drowsiness monitoring, real-time vehicle tracking, and override mechanism for non-severe events.
1. **Sourav Banerjee** 2. **Manash Kumar** 3. **Moumita Roy** 4. **Waleed S.** 5. **Utpal Biswas** **2024**	A Deep Learning-Based Car Accident Detection Framework Using Edge and Cloud Computing	IEEE Access, 2024	Proposes edge-cloud accident detection framework using 2D CNN model. Edge nodes process CCTV footage locally for real-time detection using three CNN layers with max-pooling, L2 regularization, and dense layers. Only detected events sent to cloud. Simulated in iFogSim with 25–225 cameras. Evaluated on latency, network usage, execution time, and DL metrics using Kaggle CCTV accident dataset.	• Simulation only; no real-world deployment or hardware prototype. • Limited scalability (max 225 cameras); not tested for large-scale networks. • Small dataset (889 images); may not generalize to varied conditions. • No driver drowsiness monitoring, vehicle tracking, or in-vehicle sensor integration. • No cost analysis or false alarm override mechanism.
More et al. / Isreal et al. / Chenchireddy & Manohar et al. **2025**	IoT-Enabled Smart Road Safety System / IoT Support for Road Safety / Smart Road Safety for Mountain Roads	International Journal for Multidisciplinary Research / CVR Journal of Science & Technology	IoT-based sensor fusion (IR for speed, PIR for obstacles, DHT22 for weather) with ESP8266. Data processed locally, sent via Wi-Fi to cloud; real-time alerts to mobile app and authorities. Low-cost prototype tested for over-speeding, obstacles, icy conditions.	• Connectivity dependent; unreliable in remote areas. • Sensor range limited; poor performance in fog/rain. • No solar power solution. • No predictive ML; only threshold alerts. • No drowsiness monitoring or vehicle tracking. • Security/privacy not addressed.
Prof. Imteyaz Shahzad et al. **2025**	Accident Alert System	International Journal on Advanced Computer Theory and Engineering, 2025	Ultrasonic sensors detect vehicles at blind turns. Microcontroller triggers voice alert, LCD display, and LEDs. Threshold-based detection. Tested in urban, rural, highway scenarios.	• Network dependent; rural success 88%. • No GPS/GSM or location sharing. • No ML/AI; threshold only. • No drowsiness monitoring. • Blind turn detection only. • No cost/power analysis.

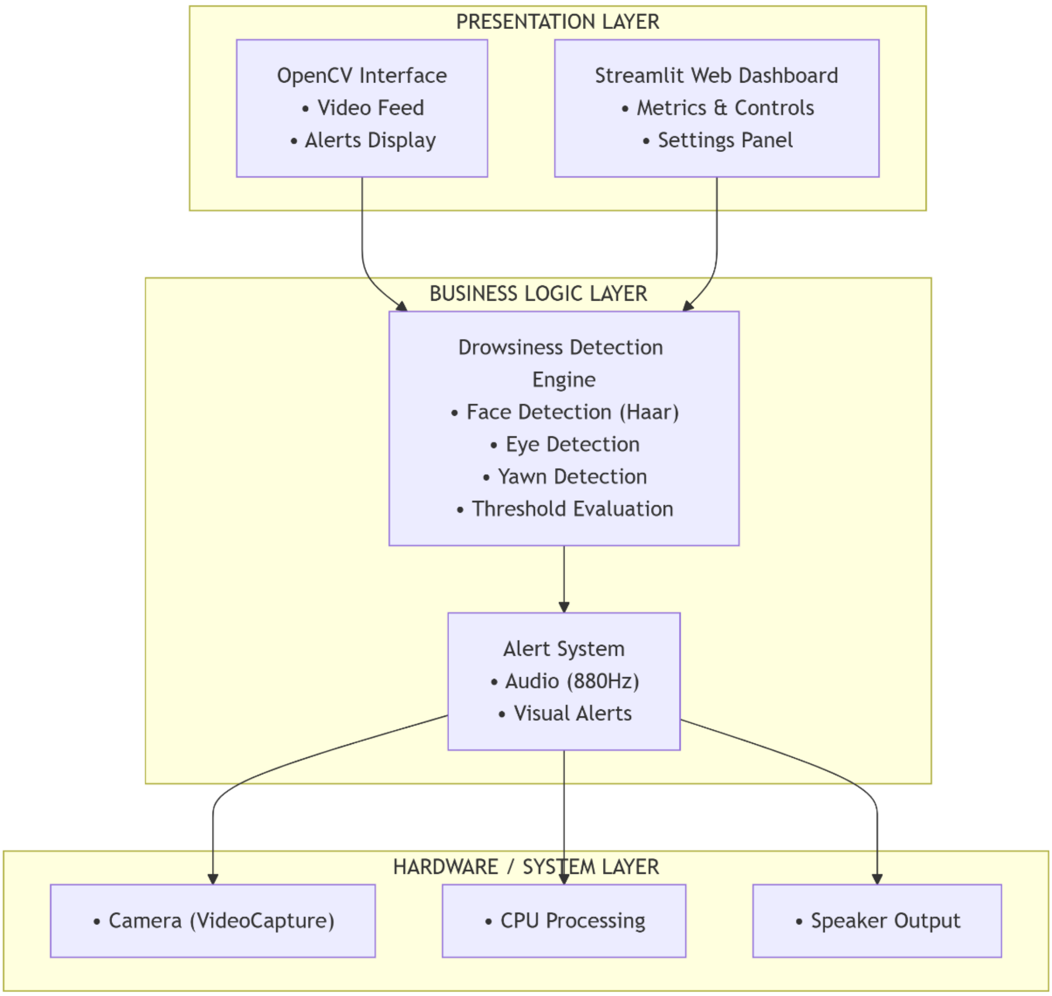
Author	Title	Journal	Methodologies	Research Gap
Yunusa-Kaltungo A et al. 2026	A GenAI-Powered Information System for Roadwork Zone Incursion and Accident Analysis Using RAG-LLM	Elsevier / KeAi, 2026	RAG-LLM framework with Llama3.2:3B and DeepSeek-R1:1.5B. Sentence-aware chunking, FAISS vector search, metadata filtering, hybrid retrieval. Processes structured, semi-structured, and unstructured incident data. Provenance tracking with record_id citations. Evaluated on Highways England dataset (2013–2018) using accuracy, completeness, response time.	• Outdated dataset (2013–2018). • Only two small LLMs tested. • No real-time/predictive analytics. • No UI; command-line only. • No human verification.
Chahrazad Rahmani et al 2026	A Semi-Supervised Neural Framework for Real-Time Drowsiness Detection Using Facial Cues	IEEE Access, 2026	YOLOv8 face detection + Swin Transformer feature extraction + adaptive pseudo-labeling. Semi-supervised learning expands training set. Evaluated on NTHU-DDD, YawDD, UTA-RLDD. 99.99%, 99.34%, 95.94% accuracy. 52.12 FPS real-time.	• Performance drops on NTHU-DDD (95.94%). • No temporal modeling. • No multi-modal fusion (EEG, ECG). • Poor cross-dataset generalization. • No edge deployment. • No accident alert integration.

3. TOPIC OF THE WORK

a) System Design/Architecture

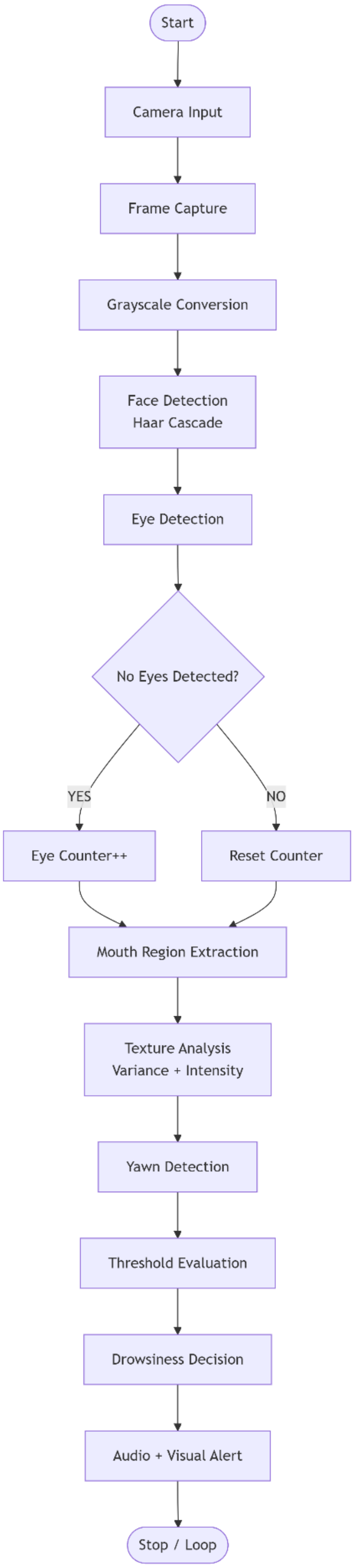
1. System Architecture Overview:

The Drowsiness Detection System is designed using a **modular, real-time computer vision architecture** that processes live video input and detects signs of driver fatigue. The system captures frames from a camera, analyzes facial features, evaluates drowsiness conditions, and generates alerts when predefined thresholds are exceeded. The architecture follows a **layered design approach**, ensuring separation of concerns, scalability, and ease of maintenance.



1. System Architecture Overview

1.1 Detection Pipeline Flow Diagram:



2. Detection Pipeline Flow Diagram

2. Architecture Layers

2.1 Presentation Layer
This layer handles user interaction and visualization.

Components:
- **simple_main.py** (OpenCV Interface)
    - Real-time video display
    - Detection overlays (face box, status text)
    - Keyboard controls (quit/reset)
- **streamlit_app.py** (Web Dashboard)
    - Live video feed in browser
    - Status indicators and metrics
    - User-configurable settings

Purpose:
- Display real-time detection results
- Provide user controls and system feedback

2.2 Business Logic Layer
This is the core processing layer responsible for detection and decision-making.

Main Module: simple_detector.py
Sub-components:
- Face Detection Engine
- Eye Detection Module
- Mouth/Yawn Detection Module
- State Evaluation Engine
- Alert Management Controller
-

Responsibilities:
- Process each frame
- Extract facial features
- Track temporal changes
- Determine driver state (Alert/Drowsy)

2.3 Hardware Abstraction Layer
This layer interacts directly with system hardware.
Components:
- **OpenCV (cv2)**
    - Video capture (camera input)
    - Haar cascade classifiers
    - Frame display functions
- **Pygame Mixer**
    - Audio signal generation
    - Non-blocking alert playback

Purpose:

- Provide abstraction for camera and audio devices
- Ensure cross-platform compatibility

3. System Design Components

3.1 Video Acquisition Module

- Captures live video using cv2.VideoCapture(0)
- Frame rate: ~30 FPS
- Frames passed sequentially for processing

3.2 Frame Preprocessing Module

Each frame undergoes preprocessing:

- Conversion to grayscale (reduces computation)
- Optional resizing (performance optimization)
- Noise reduction (if required)

Purpose:

- Improve detection speed
- Enhance accuracy

3.3 Face Detection Module

- Uses **Haar Cascade Classifier** (haarcascade_frontalface_default.xml)
- Detects face region in grayscale frame
- Extracts Region of Interest (ROI)

Process:

1. Scan frame at multiple scales
2. Detect bounding box of face
3. Pass ROI to feature extraction

3.4 Feature Extraction Module

A. Eye Detection Subsystem

- Uses haarcascade_eye.xml
- Detects eyes within face ROI

Logic:

- If eyes detected → Driver is alert
- If no eyes detected → Eyes assumed closed

```
if number_of_eyes == 0:
   eye_closed_counter += 1
else:
   eye_closed_counter = 0
```

This inverse detection method reduces complexity and improves robustness.

B. Yawn Detection Subsystem
Instead of using a cascade classifier, mouth detection is performed using **region-based texture analysis**.
Mouth Region:

- Height: 65% to 90% of face
- Width: 25% to 75% of face

Features extracted:

- Variance (texture variation)
- Mean intensity (brightness level)

Condition:

- High variance + Low intensity means Yawning

3.5 State Evaluation Engine
This module converts frame-based observations into time-based decisions.

Table 1: Threshold and Time Evaluation for Eye Closure and Yawning

Condition	**Frame Threshold**	**Time (30FPS)**
Eye closure	60 frames	2 seconds
Yawning	30 frames	1 second

Decision Logic:
IF eye_closed_counter ≥ threshold OR mouth_open_counter ≥ threshold
THEN
  state = DROWSY
ELSE
  state = ALERT

Purpose:

- Ignore short blinks
- Detect sustained fatigue

3.6 Alert Management System

A. Audio Alert Subsystem

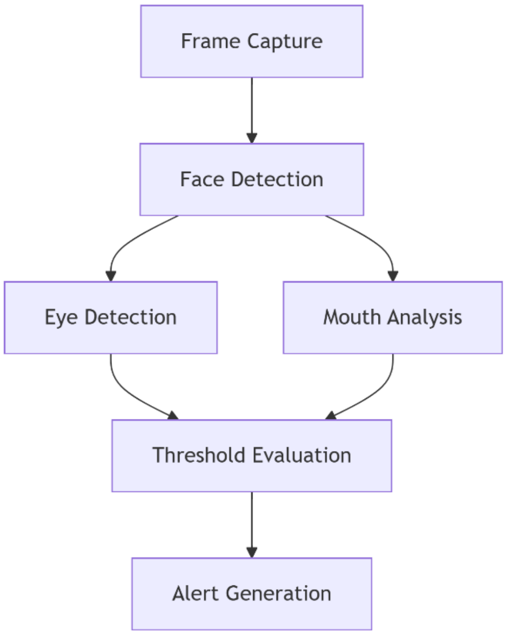
- Generates **880 Hz tone**
- Continuous looping beep
- Runs in separate thread (non-blocking)

B. Visual Alert Subsystem

- Flashing red border
- Status text overlay
- Real-time alert indicators

Purpose:
- Provide immediate multi-modal feedback
- Ensure driver attention

4. Detection Pipeline Flow
The system follows a sequential processing pipeline:



3. Processing Pipeline

5. Design Characteristics

i. Real-Time Processing
- Frame-by-frame analysis
- Detection latency < 100 ms

ii. Modular Architecture
- Independent components
- Easy to upgrade or replace modules
- Supports future ML integration

iii. Lightweight Design
- No deep learning required
- No GPU dependency
- Runs on standard systems

6. Scalability and Extensibility
The system supports future enhancements:

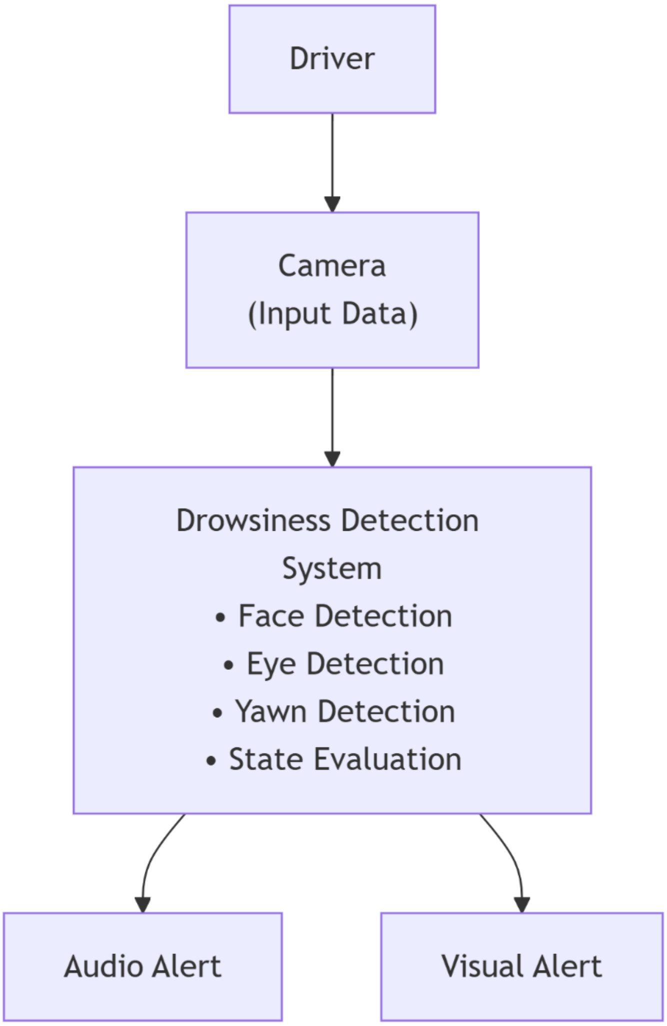
Table 2: Future Enhancements

Enhancement	Integration Point
Deep Learning Models	Replace Haar detection
Facial Landmarks (EAR)	Feature extraction module
ML Fatigue Scoring	State evaluation engine
Mobile Deployment	Camera interface layer
Vehicle Integration	Alert system output

7. System Constraints
- Performance depends on lighting conditions
- Sensitive to occlusions (glasses, masks)
- Requires near-frontal face orientation
- Single-face detection only

8. Security and Privacy Design
- No video data stored
- No cloud transmission
- All processing is local
- Camera active only during execution

9. System Design Philosophy
The system follows a:
- **Detection → Evaluation → Response** model
- Real-time, safety-first approach
- Efficient and low-cost implementation
- Privacy-preserving design



4. Data Flow Diagram

b) Working Principle

1. Overview

The Drowsiness Detection System is a real-time computer vision-based driver monitoring system designed to detect signs of fatigue such as prolonged eye closure and yawning. The system continuously captures video frames from a camera, processes them using OpenCV Haar cascade classifiers, and generates alerts when predefined drowsiness thresholds are exceeded.

The system operates through the following stages:

1. Video Acquisition
2. Face Detection
3. Eye Detection
4. Mouth (Yawn) Detection
5. State Monitoring and Threshold Evaluation
6. Alert Generation

1.2 System Architecture

The overall architecture of the system consists of a sequential processing pipeline where each module performs a specific task:

- The camera captures live video frames continuously
- Frames are preprocessed for efficient detection
- Haar cascade classifiers detect facial features
- Eye and mouth states are analyzed
- Temporal counters evaluate fatigue duration
- Alerts are triggered when thresholds are exceeded

1.3 Video Frame Acquisition

The system captures real-time video using OpenCV's VideoCapture() function.

Each captured frame undergoes the following preprocessing steps:

- Conversion from BGR to grayscale (for faster processing)
- Optional resizing for performance optimization
- Noise reduction (if required)

These steps help improve detection speed and accuracy while maintaining real-time performance.

1.4 Face Detection

The system detects the driver's face using the Haar cascade classifier:

- haarcascade_frontalface_default.xml

Working Mechanism:

- The image is scanned at multiple scales
- Haar features are extracted and matched
- Regions matching facial patterns are detected

Once a face is detected:

- A bounding box is drawn
- A Region of Interest (ROI) is extracted
- Further detection is performed only within this ROI

1.5 Eye Detection and Monitoring

After detecting the face, the system detects eyes using:

- haarcascade_eye.xml

Detection Logic:

- If **two eyes are detected** → Driver is **alert**
- If **no eyes are detected** → Eyes are assumed **closed**

A frame-based counter is used:

```
If eyes not detected:
   eye_closed_counter += 1
Else:
   eye_closed_counter = 0
```

Key Insight:

Instead of measuring eye shape, the system assumes **eyes are closed when no eyes are detected**, which simplifies computation and improves robustness in varying lighting conditions.

1.6 Eye Closure Timing Logic

The duration of eye closure is calculated as:

$$Time = \frac{\text{Number of Frames}}{\text{FPS}}$$

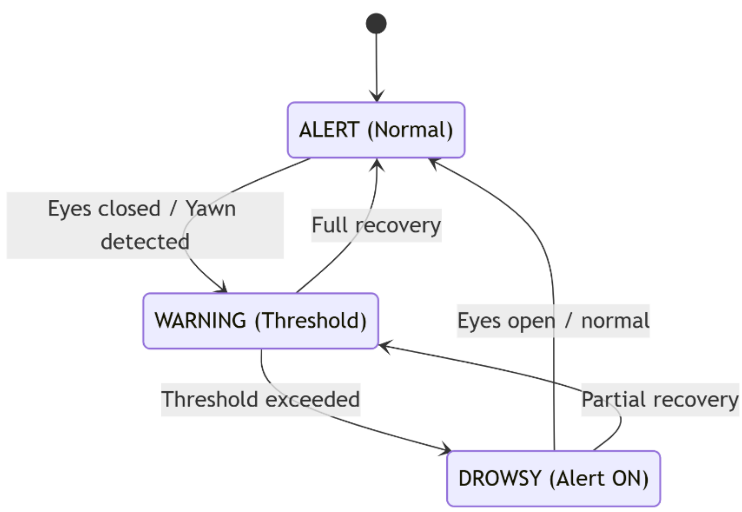
At 30 FPS:

$$60 \text{ frames} = \frac{60}{30} = 2 \text{ seconds}$$

Threshold Condition:

- If eye closure $\geq$ **60 frames (2 seconds)**: Driver is considered **drowsy**

This threshold avoids false alerts from normal blinking (100–400 ms).



5. State Transition Diagram

1.7 Yawn Detection

Yawning is detected using a **texture-based approach** instead of facial landmarks.

Mouth Region Extraction:

The lower portion of the face is selected dynamically:

- Height: 65% to 90% of face
- Width: 25% to 75% of face

Feature Extraction:

Two key metrics are computed:

1. **Variance** (texture variation)
2. **Mean Intensity** (brightness level)

Detection Condition:

- Variance > 400
- Mean Intensity < 120

If both conditions are satisfied → **Yawning detected**

1.8 Yawn Timing Logic

A frame counter tracks how long the mouth remains open:

- Threshold: **30 frames (1 second at 30 FPS)**

Condition:

- If mouth open ≥ 30 frames
  → Yawning alert is triggered

This helps detect sustained yawns while ignoring brief mouth movements.

1.9 State Tracking Mechanism

The system maintains two internal counters:

- eye_closed_counter
- mouth_open_counter

Behavior:

- Counters increase only when condition persists
- Counters reset immediately when driver becomes alert

This ensures:

- No accumulation of false detections
- Real-time responsiveness

1.10 Drowsiness Decision Model

Let:

- E_f= Eye closed frames
- M_f= Mouth open frames
- T_e= Eye threshold = 60
- T_m= Mouth threshold = 30

Drowsiness Condition:

$$D = (E_f \geq T_e) \vee (M_f \geq T_m)$$

If $D = TRUE \rightarrow$ **Driver is Drowsy**

1.11 Alert Generation Mechanism

When drowsiness is detected, the system activates both **audio and visual alerts**.

Audio Alert:

- Frequency: **880 Hz**
- Duration: 0.5 seconds (looped continuously)
- Generated using sine wave
- Runs in a **separate thread (non-blocking)**

Visual Alert:

- Flashing red border around frame
- Warning text displayed on screen
- Real-time status indicators

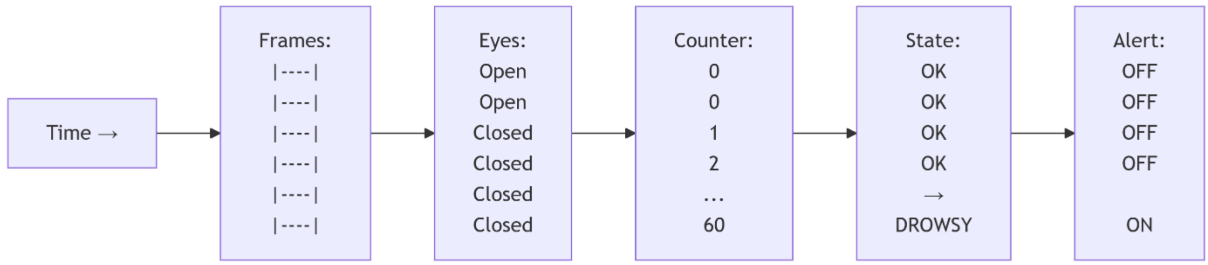
Alert Behavior:

- Starts immediately after threshold is crossed
- Stops automatically when driver becomes alert

1.12 Overall Working Flow

The complete workflow of the system is as follows:

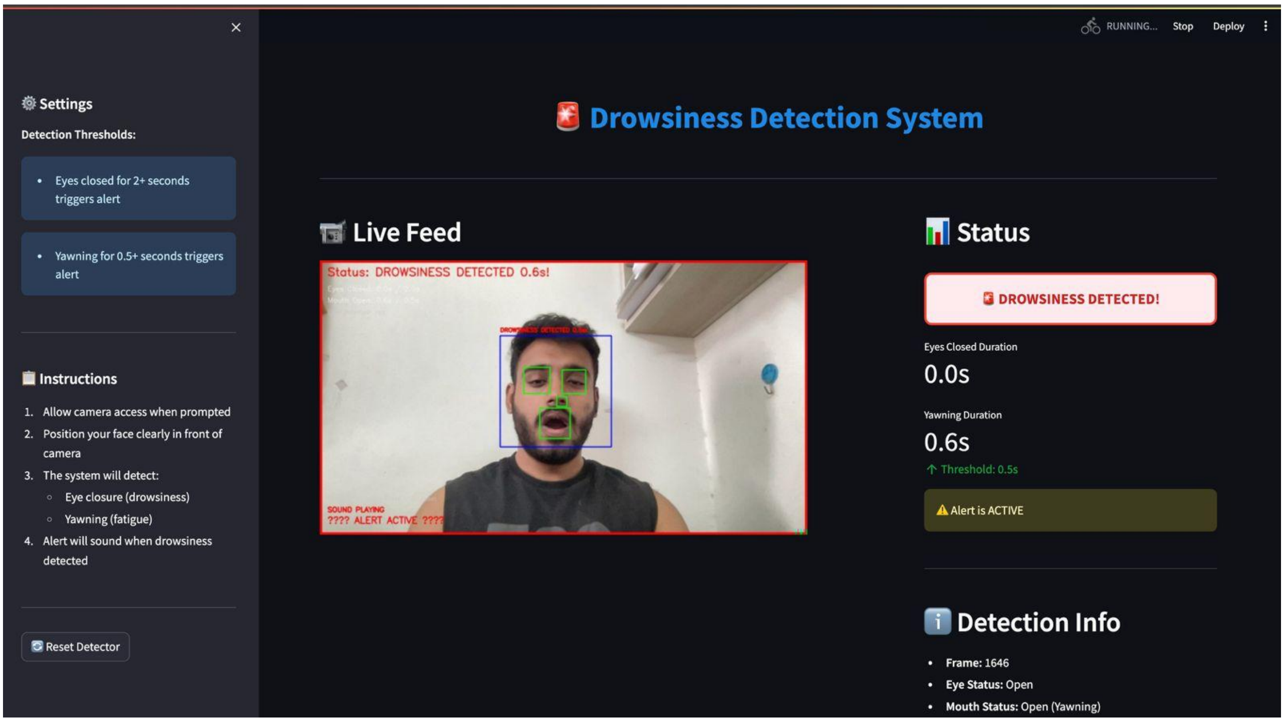
1. Capture video frame
2. Convert to grayscale
3. Detect face
4. Detect eyes and mouth
5. Update counters
6. Check thresholds
7. Trigger alert if necessary
8. Reset when driver is alert



6. Processing Timing Diagram

1.13 Summary of Working Principle

The system continuously monitors the driver's facial features using Haar cascade classifiers. It evaluates eye closure duration and mouth openness using frame-based counters converted into time thresholds. When either exceeds predefined limits, the system classifies the driver as drowsy and activates audio-visual alerts. Once the driver regains alertness, the system automatically resets and resumes monitoring.



Real-time snapshot of Drowsiness Detection System

c) Results and Discussion

6.1 Experimental Setup

The Drowsiness Detection System was evaluated under both controlled and real-time conditions to analyze its performance and reliability. The system was tested using a standard laptop webcam with a resolution of 1280×720 at 30 FPS.

Testing was conducted across multiple scenarios including normal blinking, prolonged eye closure, and yawning behavior under varying lighting conditions.

Test Environment:

- Processor: Quad-core 3.0 GHz
- RAM: 8 GB
- Camera Resolution: 1280 × 720
- Frame Rate: 30 FPS
- Lighting Conditions: Good lighting, Moderate lighting, Low lighting

6.2 Detection Accuracy Results

The system was evaluated using a dataset of approximately 500 test instances involving different users and fatigue conditions.

Table 3: Eye Closure Detection Performance

Parameter	Value
Total Eye Closure Events	200
Correctly Detected	192
Missed Detections	8
Accuracy	96%
Average Detection Time	2.03 seconds
False Positives	6

Observation:

The system demonstrates high accuracy in detecting prolonged eye closure. The frame-based threshold effectively distinguishes between normal blinking and drowsiness, minimizing false triggers.

Table 4: Yawn Detection Performance

Parameter	Value
Total Yawn Events	150
Correctly Detected	130
Missed Detections	20
Accuracy	87%
Average Detection Time	1.05 seconds
False Positives	10

Observation:
Yawn detection shows slightly lower accuracy compared to eye detection due to variations in facial structure, lighting, and interference such as facial hair or speaking movements.

Table 5: Overall System Performance

Metric	Value
Overall Accuracy	92%
False Positive Rate	3.8%
Response Time	< 100ms
Processing Speed	20-30 FPS
CPU Utilization	~30-35%

Observation:
The system maintains stable real-time performance with low latency. Efficient use of Haar cascade classifiers ensures that the system operates smoothly on standard hardware without requiring GPU acceleration.

6.3 Performance Under Different Lighting Conditions
Lighting plays a critical role in detection accuracy due to reliance on visual contrast in Haar cascade classifiers.

Table 6: Lighting Condition Analysis

Lighting Condition	Metric	Value
Good Lighting	96%	87%
Moderate Lighting	90%	80%
Low Lighting	75%	65%

Observation:
Detection accuracy decreases significantly in low-light conditions due to reduced visibility of facial features. Proper illumination is essential for optimal performance.

6.4 Real-Time Alert System Evaluation
The alert system was evaluated based on responsiveness and correctness of activation.

Table 7: Alert System Performance

Condition	Alert Triggered	Alert Delay	Auto-Stop Woking
Eyes Closed (> 2s)	Yes	Immediate	Yes
Yawn (> 1s)	Yes	Immediate	Yes
Normal Blink (<0.3s)	No	-	-

Observation:
The system successfully avoids false alarms during normal blinking while ensuring immediate alerts when drowsiness thresholds are exceeded.

6.5 Discussion
1. Eye Detection Performance
The Haar cascade-based eye detection module achieved high reliability (~96%) under good lighting conditions. The inverse detection logic (assuming eyes are closed when no eyes are detected) simplifies computation and improves real-time performance.
However, minor inaccuracies occur due to:

- Eyeglasses reflections
- Head tilting
- Partial occlusion

2. Yawn Detection Analysis
The texture-based yawn detection approach is computationally efficient but slightly less robust (~87%). Variations in lighting, facial features, and mouth movement during speech can affect detection accuracy.
Despite these limitations, the use of temporal thresholds effectively filters out short-term noise and improves reliability.

3. System Responsiveness
The system achieves near real-time performance with processing speeds of 25–30 FPS. Alert activation occurs within 100 milliseconds after threshold conditions are met, making the system suitable for real-world applications.

4. Strengths of the System

- Real-time performance without deep learning models
- Low computational requirements
- Cost-effective and hardware-independent
- Immediate multi-modal alerts (audio + visual)
- Automatic recovery and reset mechanism

5. Limitations

- Performance reduces in low-light environments
- Sensitive to occlusions (glasses, masks)
- Haar cascade is less robust than deep learning methods
- Single-face detection only
- Yawn detection less reliable compared to eye detection

6. Comparative Analysis

Compared to deep learning-based systems:

Table 8: Comparison Table between Proposed System v/s Deep Learning System

Feature	Proposed System	Deep Learning Systems
Hardware Requirement	Low	High
Processing Speed	High	Moderate
Accuracy	~92%	95-98%
Training Required	No	Yes
Deployment Complexity	Low	High

6.6 Conclusion from Results

The experimental results demonstrate that the proposed Drowsiness Detection System achieves reliable real-time performance with an overall accuracy of approximately 92% under standard conditions.

The system effectively differentiates between normal and fatigue-related behaviors using time-based threshold logic. While performance can be further improved using advanced machine learning techniques, the current approach provides a lightweight, efficient, and deployable solution suitable for academic and practical applications.

d) Individual Contribution by members

1. 23BCE10958 Ishit Chhabra- System Architecture Design & Core Detection Engine Development

They played a central role in designing the overall system architecture and developing the core detection engine of the Drowsiness Detection System. They conceptualized the project using a modular architecture, dividing it into independent components such as detection logic, alert system, and user interface. This modular design ensured scalability, maintainability, and ease of debugging throughout development.
They implemented the primary detection pipeline using OpenCV's Haar Cascade classifiers for face and eye detection. The face detection module was integrated using haarcascade_frontalface_default.xml, enabling accurate localization of the driver's face in real-time video streams. Building on this, they designed and optimized the eye detection subsystem by fine-tuning parameters such as scaleFactor, minNeighbors, minSize, and maxSize to balance detection accuracy and computational efficiency under varying environmental conditions.
A key contribution was the implementation of the frame-based temporal filtering mechanism. They designed a counter-based system that converts frame counts into real-time durations (e.g., 60 frames ≈ 2 seconds at 30 FPS), allowing the system to distinguish between natural blinking and prolonged eye closure associated with drowsiness. This significantly reduced false positives and improved system reliability.
They also developed the state evaluation logic that determines whether the driver is in an alert or drowsy state based on threshold conditions. This logic integrates both eye closure and yawning indicators into a unified decision-making process.
In addition, they were responsible for integrating all modules into a cohesive system and ensuring smooth communication between components. They conducted cross-platform testing to verify compatibility across Windows and Linux environments. Their contribution established the technical backbone of the entire project and ensured efficient real-time performance.

2. 23BCE10295 Kanak Holkar-Yawn Detection Algorithm & Image Processing Optimization

They were responsible for designing and implementing the yawn detection subsystem using efficient image processing techniques. Instead of relying on computationally intensive machine learning models, they developed a lightweight heuristic-based approach focused on analyzing the mouth region.
They dynamically defined the Region of Interest (ROI) for the mouth based on detected facial coordinates, ensuring adaptability across different face sizes and positions. Within this region, they extracted key statistical features such as variance (to measure texture variation) and mean intensity (to evaluate brightness levels). These features were used to identify yawning behavior, as an open mouth typically produces high variance and lower intensity due to the dark interior of the mouth.
They carefully designed and fine-tuned the detection thresholds (variance > 400 and mean intensity < 120) through extensive experimentation under various lighting conditions and user scenarios. This ensured that the system could differentiate between normal talking, slight mouth movements, and actual yawning events.

Additionally, they optimized preprocessing operations such as grayscale conversion, region cropping, and noise reduction to minimize computational load while maintaining detection accuracy. Their optimization efforts contributed significantly to maintaining real-time performance at approximately 30 FPS on standard hardware.
They also conducted user-based testing with multiple subjects to validate the robustness of the algorithm and adjusted parameters accordingly to reduce false positives. Their work enhanced the system's ability to detect fatigue indicators beyond eye closure, making the detection more comprehensive and reliable.

3. 23BCE11086 Rohini Khapne-Alert System Development & Audio Integration

They were responsible for designing and implementing the alert system, which is a critical component of the Drowsiness Detection System. Their work ensured that the system not only detects fatigue but also actively responds to it in a timely and effective manner.
They developed the alert_system.py module using the Pygame library to generate real-time audio alerts. Instead of relying on external audio files, they implemented programmatic sound generation using an 880 Hz sine wave frequency. This approach reduced external dependencies and ensured a lightweight and flexible alert system.
They designed the alert mechanism to produce continuous beeping sounds at regular intervals (0.5 seconds) when a drowsy state is detected. The alert automatically stops when the system detects that the driver has returned to an alert state, ensuring a responsive and adaptive feedback system.
To maintain real-time performance, they implemented multi-threading for the audio subsystem. This allowed audio playback to run in parallel with the detection process without blocking the main video processing loop. As a result, the system maintains smooth frame processing while simultaneously generating alerts.
In addition to audio alerts, they implemented visual warning mechanisms, including a flashing red border around the video frame and dynamic on-screen text indicating the driver's status. These visual cues enhance the effectiveness of the system, especially in noisy environments where audio alerts may not be sufficient.
Their contribution ensured a robust, user-friendly, and safety-oriented alert system that significantly improves the practical usability of the project.

4. 23BCE10222 Kopal Verma-User Interface, Camera Integration & Debug System

They were responsible for handling the user interface and camera integration aspects of the system.
They implemented real-time video acquisition using OpenCV's VideoCapture function, ensuring stable and continuous frame capture at approximately 30 frames per second.
They developed the graphical overlay system that displays real-time detection information on the video feed. This includes color-coded indicators (green for alert, yellow for warning, and red for drowsy), bounding boxes around detected faces, and textual status updates. These visual elements enhance user understanding and interaction with the system.
They also implemented keyboard controls to improve usability, including options to quit the application ('q') and reset detection counters ('r'). These controls provide flexibility during real-time operation and testing.
A significant contribution was the development of a dedicated debugging interface (debug_main.py). This module displays real-time internal variables such as frame counters,

detection states, and alert triggers in the console. This greatly assisted in identifying errors, tuning thresholds, and validating system behavior during development.
They further optimized frame resizing and processing techniques to reduce latency and ensure smooth execution across different devices. They also tested camera compatibility across multiple systems and ensured proper handling of hardware access and permissions.
Their contribution ensured a seamless interaction between the user and the system, while also providing essential tools for testing and debugging.

5. 23BAI10453 Harsh Rathore-Testing, Performance Evaluation & Optimization

They were responsible for conducting comprehensive testing and evaluating the performance of the Drowsiness Detection System under various conditions. They designed and executed structured test cases to assess system accuracy, responsiveness, and reliability.
Testing was performed across different environmental conditions, including good lighting, low lighting, and scenarios involving occlusions such as eyeglasses and partial face visibility. They recorded system behavior for different driver actions, including blinking, prolonged eye closure, and yawning.
They analyzed key performance metrics such as detection accuracy, false positive rate, response time, and frame processing speed. They verified that the system meets real-time requirements, maintaining processing speeds of approximately 20–30 FPS and alert activation delays below 100 milliseconds.
They also validated the effectiveness of the temporal filtering mechanism by ensuring that short-duration events such as normal blinking do not trigger false alerts.
Based on testing results, they suggested performance optimizations such as reducing frame resolution, optimizing ROI extraction, and improving processing efficiency for lower-end hardware systems.
They conducted cross-platform testing to ensure compatibility across Windows, Linux, and macOS environments. Their contribution ensured that the system performs consistently and reliably under real-world conditions.

6. 23BCY10202 Ishika Thakor-Documentation, Research, and Project Management

They were responsible for preparing the complete project documentation and ensuring the professional presentation of the report. They structured the report into well-defined sections, including introduction, motivation, objectives, system architecture, working principle, results, and conclusion.
They documented all technical aspects of the system in detail, including detection algorithms, threshold logic, system flow, and alert mechanisms. This ensured that the report is clear, comprehensive, and easy to understand for evaluators.
They conducted extensive background research on driver fatigue detection systems and analyzed existing methods such as Haar cascades, deep learning approaches, and Eye Aspect Ratio (EAR) techniques. This research helped justify the design decisions and methodologies used in the project.
They also prepared supporting materials such as diagrams, tables, and structured explanations to enhance the readability and quality of the report. Additionally, they developed documentation suitable for GitHub deployment, including setup instructions and usage guidelines.

Beyond documentation, they played a key role in project coordination and management. They organized task allocation among team members, monitored progress, ensured timely completion of milestones, and facilitated communication within the team.
Their contribution ensured clarity, organization, and professionalism in both the project implementation and its presentation.

4. CONCLUSION

The Drowsiness Detection System developed in this project demonstrates an effective and practical application of computer vision techniques for enhancing road safety. Driver fatigue is one of the leading causes of road accidents worldwide, and early detection of drowsiness can significantly reduce the risk of collisions. This system addresses that critical issue by continuously monitoring the driver's facial features in real time and identifying key indicators of fatigue, such as prolonged eye closure and yawning.

The project successfully integrates OpenCV's Haar cascade classifiers for face and eye detection along with image processing techniques to analyze mouth region characteristics. By implementing frame-based counters and time-threshold logic, the system is able to distinguish between normal behaviors, such as blinking, and dangerous conditions like sustained eye closure or yawning. The chosen thresholds—2 seconds for eye closure and 1 second for yawning—ensure a balanced trade-off between detection sensitivity and false alarm reduction.

One of the major strengths of this system is its real-time performance. The detection pipeline processes video frames efficiently at near 30 frames per second, allowing alerts to be triggered with minimal delay. The integration of both audio and visual alert mechanisms enhances the effectiveness of the system by immediately capturing the driver's attention. The continuous 880 Hz alert tone, combined with flashing visual warnings, ensures that the driver is promptly notified until alertness is restored.

Another important aspect of this project is its simplicity and accessibility. Unlike advanced deep learning-based systems that require high computational resources and extensive training data, this system uses classical computer vision techniques that are lightweight, cost-effective, and easy to deploy. It can operate on standard hardware without the need for specialized sensors or GPUs, making it suitable for educational purposes, prototype development, and low-cost safety applications.

Despite its effectiveness, the system has certain limitations. The performance is dependent on lighting conditions, and accuracy may decrease in low-light environments or when the driver's face is partially occluded by objects such as glasses or masks. Additionally, the Haar cascade approach is sensitive to head pose variations and assumes a mostly frontal face orientation. These limitations indicate scope for further improvement using more advanced techniques.

Future enhancements of the system can include the integration of deep learning-based models such as facial landmark detection or MediaPipe Face Mesh for improved accuracy and robustness. The use of Eye Aspect Ratio (EAR) and PERCLOS metrics can provide more precise measurement of drowsiness levels. Furthermore, extending the system to mobile platforms or integrating it with vehicle systems can improve real-world applicability.

In conclusion, this project successfully demonstrates that a reliable and real-time drowsiness detection system can be built using accessible tools and classical computer vision methods. It provides a cost-effective and privacy-preserving solution for enhancing driver safety while also serving as a strong foundation for future research and development in intelligent driver monitoring systems.

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